

# 120 Days of Quantitative Finance

## Month 3: Machine Learning for Finance

### Week 11: Portfolio Optimization and Risk Management using ML

#### Day 75: Risk Management with Machine Learning

## 1 Introduction

Risk management is one of the most important components of financial decision making. While portfolio optimization focuses on maximizing returns, risk management aims to control potential losses and ensure portfolio stability.

Traditional risk management techniques rely heavily on statistical models such as variance, Value-at-Risk (VaR), and Expected Shortfall.

However, financial markets are complex, nonlinear, and often non-stationary. Machine learning methods provide flexible tools that can improve risk estimation and prediction.

## 2 Key Risk Measures

Several quantitative metrics are commonly used to measure financial risk.

### 2.1 Volatility

Volatility measures the variability of asset returns and is often used as a proxy for risk.

$$\sigma = \sqrt{\frac{1}{n} \sum_{t=1}^n (r_t - \bar{r})^2}$$

where  $r_t$  represents asset returns and  $\bar{r}$  is the average return.

## 2.2 Value-at-Risk (VaR)

Value-at-Risk estimates the maximum expected loss over a given time horizon at a specified confidence level.

$$P(L > VaR_\alpha) = 1 - \alpha$$

where  $\alpha$  represents the confidence level.

## 2.3 Expected Shortfall

Expected Shortfall measures the expected loss conditional on losses exceeding VaR.

$$ES_\alpha = E[L|L > VaR_\alpha]$$

Expected Shortfall captures tail risk more effectively than VaR.

# 3 Limitations of Traditional Risk Models

Traditional risk models often rely on simplifying assumptions such as:

- Normally distributed returns
- Linear relationships between variables
- Stable statistical properties

In reality, financial markets frequently exhibit:

- heavy-tailed return distributions
- volatility clustering
- structural breaks
- regime changes

These features reduce the effectiveness of traditional models.

## 4 Machine Learning in Risk Management

Machine learning models can capture nonlinear relationships and complex interactions in financial data.

Applications of machine learning in risk management include:

- Volatility forecasting
- Tail risk estimation
- Market crash prediction
- Dynamic portfolio risk monitoring

## 5 ML-Based Risk Prediction

Machine learning models can be used to estimate risk metrics by learning patterns from historical data.

Suppose we have a set of features  $X_t$  describing market conditions at time  $t$ . A machine learning model can estimate future volatility or risk:

$$\hat{\sigma}_{t+1} = f(X_t)$$

where  $f(\cdot)$  represents a machine learning model.

Common models used include:

- Random Forests
- Gradient Boosting
- Neural Networks

## 6 Dynamic Risk Control

Once risk predictions are obtained, portfolio exposure can be adjusted dynamically.

For example, portfolio leverage can be scaled according to predicted volatility:

$$w_t = \frac{w_0}{\hat{\sigma}_t}$$

This reduces exposure during high-risk periods and increases exposure when risk is lower.

## 7 Conclusion

Machine learning offers powerful tools for improving risk estimation and management in financial markets.

By combining predictive models with traditional risk metrics, investors can build more adaptive and resilient portfolio strategies.